

TASK 2 – STUDENT PROFILE WEB APPLICATION

JavaScript Classes, Objects and DOM Manipulation

Task Name:	Student Profile using JavaScript	File Name:	index.html
Repository:	full-stack-web-development	GitHub Folder:	01-Observation/obsTask2_student_profile

1. AIM

To create a dynamic Student Profile web application using a JavaScript class, object creation, DOM manipulation, event handling, and basic CSS styling.

2. TASK STATEMENT

Create a JavaScript Student class with properties Name, Roll Number, Department and CGPA. Take values from the user, create a Student object, and dynamically generate the student profile on the webpage when the button is clicked. The application also validates CGPA so that it is between 3 and 10 with up to two decimal places.

3. CONCEPTS DEMONSTRATED

- JavaScript classes and objects
- DOM selection using `document.getElementById()`
- Creating HTML elements using `document.createElement()`
- Modifying content using `textContent`
- Adding elements using `appendChild()`
- Event handling using `addEventListener()`
- Input validation using `parseFloat()`, `isNaN()`, range checks and a regular expression
- Basic CSS styling of the generated profile

4. PROGRAM WORKING

- The user enters Name, Roll Number, Department and CGPA.
- The Create Profile button is selected from the DOM and a click event is attached.
- The entered values are obtained from input elements and CGPA is validated (3–10 range, max 2 decimals).
- A Student object is created with validated values.
- Previous profile content is cleared, and new HTML elements are dynamically created.
- The Student object's properties are assigned via `textContent` and appended to the profile container.

5. CGPA VALIDATION

The application accepts CGPA values from 3 through 10. The regular expression `/^\d+(\.\d{1,2})?$/` allows integers or decimals with up to two decimal places.

Input	Result
2.99	Invalid
3	Valid
8.42	Valid
10	Valid
10.01	Invalid
8.123	Invalid

6. COMPLETE SOURCE CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Profile</title>
  <style>
    body { font-family: Arial; text-align: center; background-color: #f2f2f2; }
    input { width: 250px; padding: 8px; margin: 5px; box-sizing: border-box; }
    button { padding: 10px 20px; margin: 10px; cursor: pointer; }
    #profile { width: 350px; margin: 20px auto; padding: 20px; background-color: white; border: 2px solid #333;
border-radius: 10px; text-align: left; }
    #profile h2 { text-align: center; }
  </style>
</head>
```

```

<body>
<h1>Student Profile</h1>
<input type="text" id="name" placeholder="Enter Name"><br>
<input type="text" id="roll" placeholder="Enter Roll Number"><br>
<input type="text" id="department" placeholder="Enter Department"><br>
<input type="number" id="cgpa" placeholder="Enter CGPA" min="3" max="10" step="0.01"><br>
<button id="createBtn">Create Profile</button>
<div id="profile"></div>
<script>
class Student {
  constructor(name, roll, department, cgpa) {
    this.name = name;
    this.roll = roll;
    this.department = department;
    this.cgpa = cgpa;
  }
}

const button = document.getElementById("createBtn");
button.addEventListener("click", function() {
  const name = document.getElementById("name").value;
  const roll = document.getElementById("roll").value;
  const department = document.getElementById("department").value;
  const cgpa = document.getElementById("cgpa").value;

  const cgpaValue = parseFloat(cgpa);
  if (isNaN(cgpaValue) || cgpaValue < 3 || cgpaValue > 10 || !/^\\d+(\\.\\d{1,2})?$/\\.test(cgpa)) {
    alert("CGPA must be between 3 and 10 with up to 2 decimal places.");
    return;
  }

  const student = new Student(name, roll, department, cgpa);
  const profile = document.getElementById("profile");
  profile.innerHTML = "";

  const heading = document.createElement("h2");
  const namePara = document.createElement("p");
  const rollPara = document.createElement("p");
  const departmentPara = document.createElement("p");
  const cgpaPara = document.createElement("p");

  heading.textContent = "Student Profile";
  namePara.textContent = "Name : " + student.name;
  rollPara.textContent = "Roll No : " + student.roll;
  departmentPara.textContent = "Department : " + student.department;
  cgpaPara.textContent = "CGPA : " + student.cgpa;

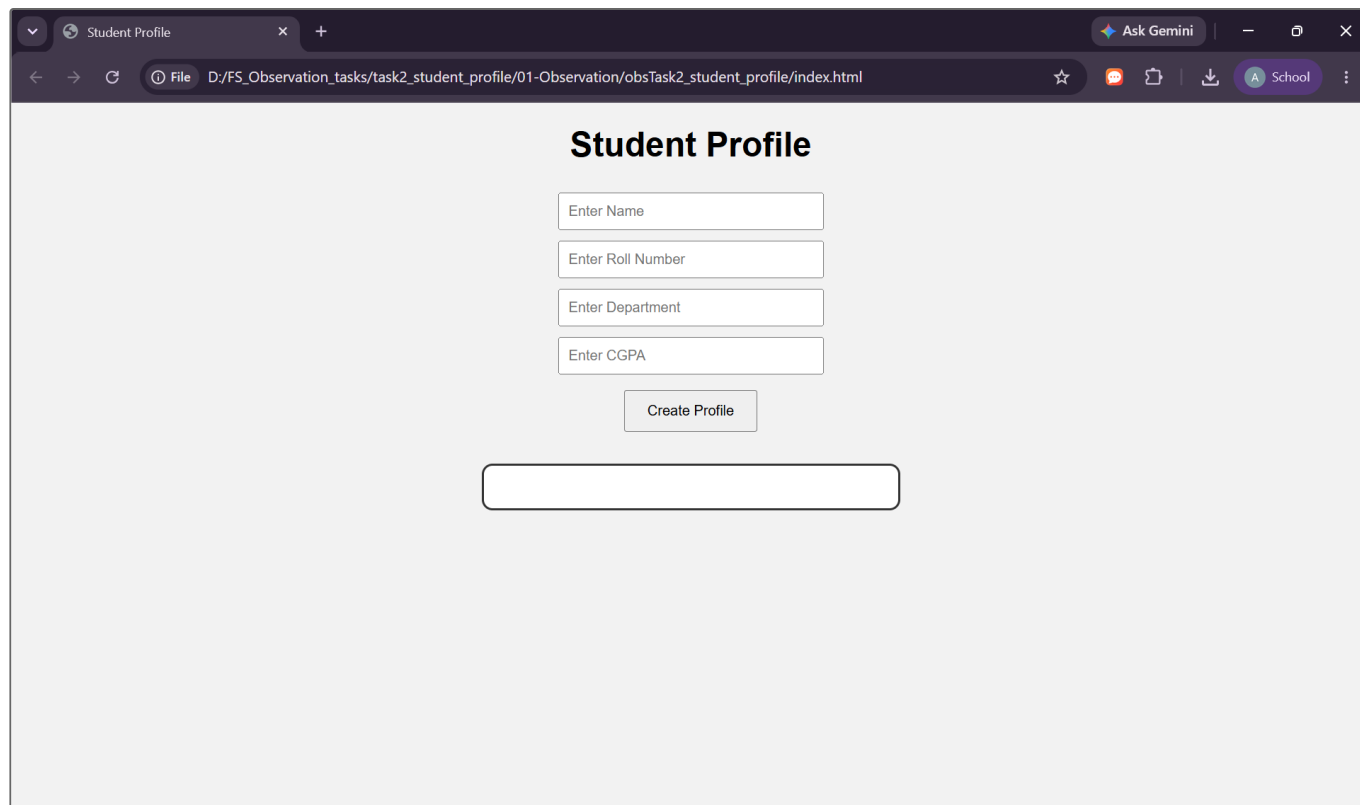
  profile.appendChild(heading);
  profile.appendChild(namePara);
  profile.appendChild(rollPara);
  profile.appendChild(departmentPara);
  profile.appendChild(cgpaPara);
});
</script>
</body>
</html>

```

7. EXPECTED OUTPUT & EXECUTION SCREENSHOTS

Screenshot 1: Initial Application Form Interface

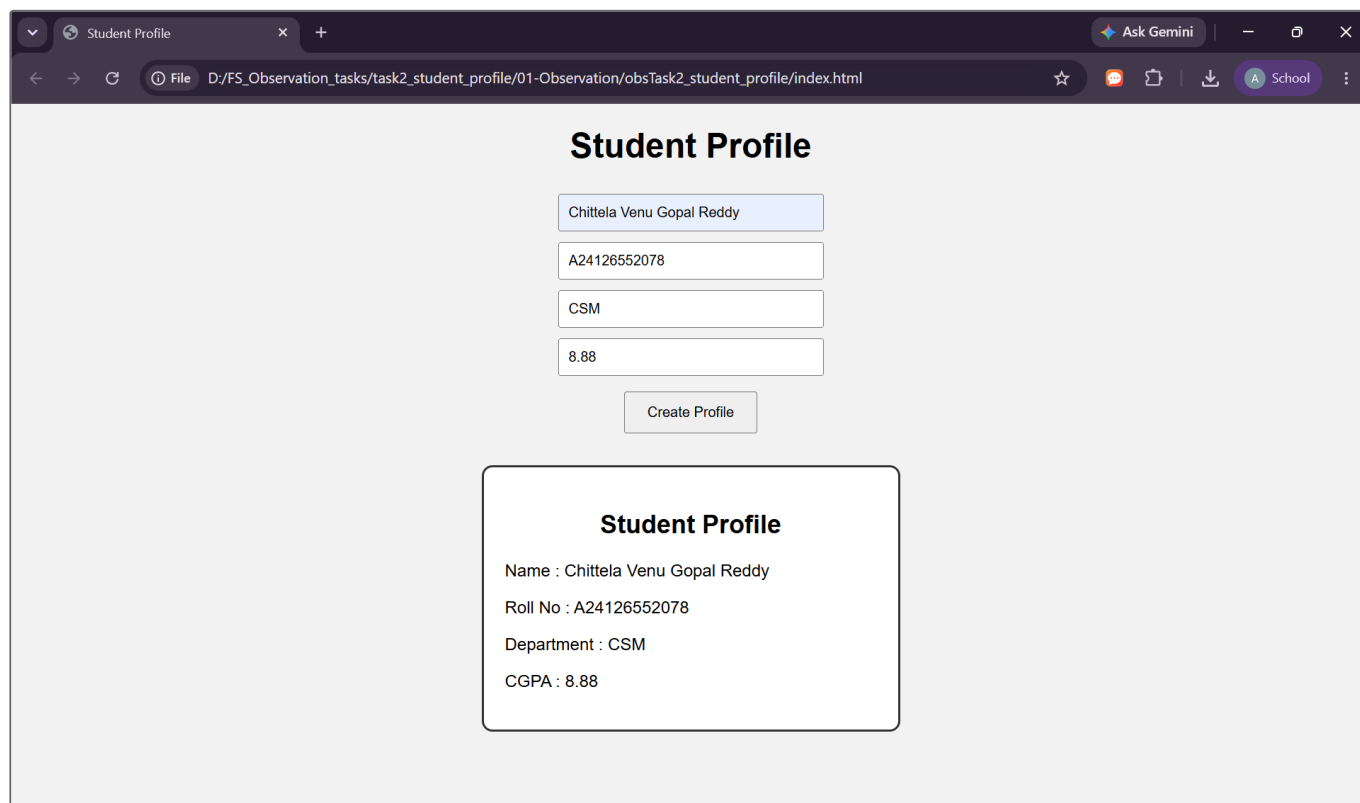
Initial web page layout with input fields for Name, Roll Number, Department, and CGPA alongside the submit button.



The screenshot shows a web browser window with the title "Student Profile". The address bar displays the file path: "D:/FS_Observation_tasks/task2_student_profile/01-Observation/obsTask2_student_profile/index.html". The page content features a heading "Student Profile" followed by four input fields: "Enter Name", "Enter Roll Number", "Enter Department", and "Enter CGPA". Below these fields is a "Create Profile" button. At the bottom of the form area, there is a large, empty rectangular box.

Screenshot 2: Generated Student Profile Output

Form submitted with valid student details and the dynamically rendered profile card on the webpage.

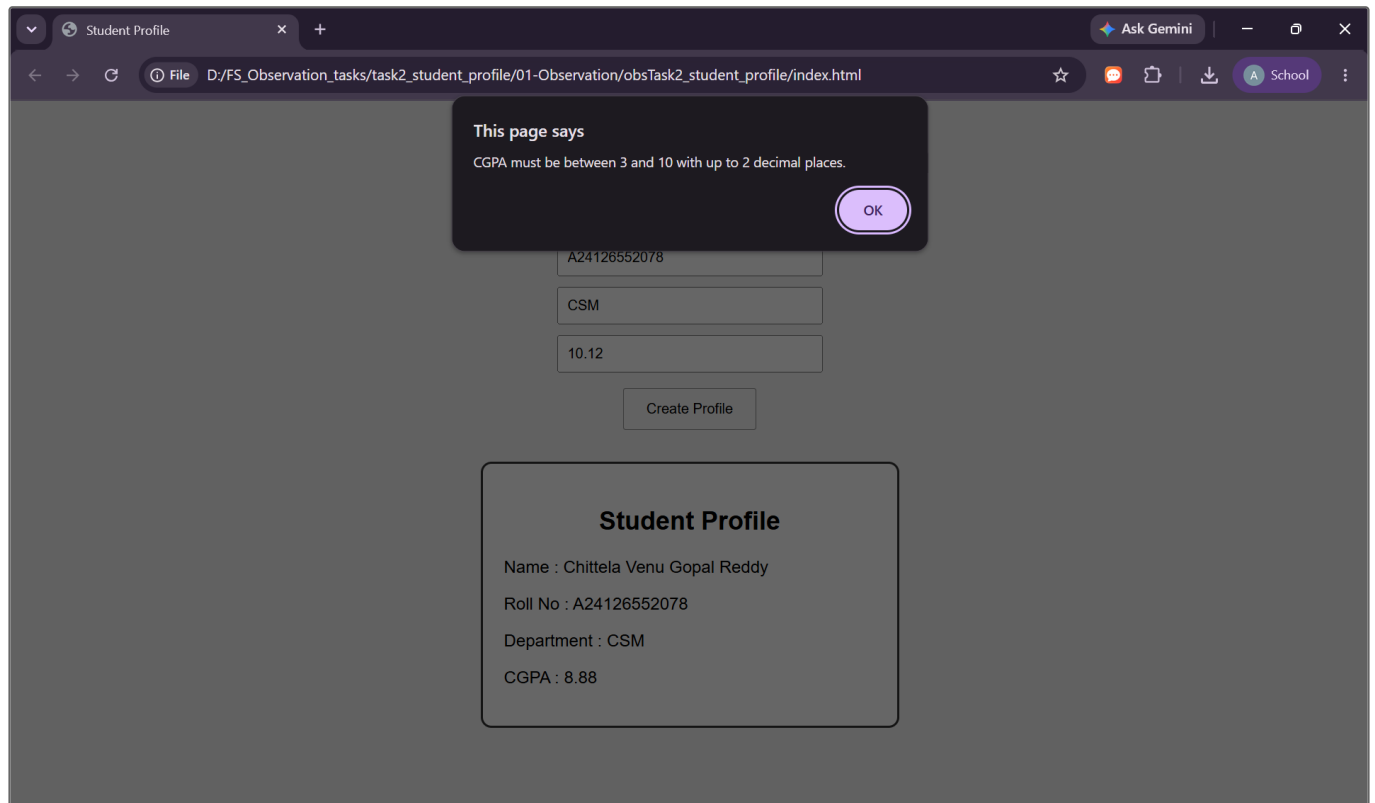


The screenshot shows the same web browser window as Screenshot 1, but with the form fields populated with student details. The "Enter Name" field now contains "Chittela Venu Gopal Reddy", "Enter Roll Number" contains "A24126552078", "Enter Department" contains "CSM", and "Enter CGPA" contains "8.88". The "Create Profile" button is still present. Below the form fields, a new "Student Profile" card has been dynamically rendered, displaying the following information:

- Name : Chittela Venu Gopal Reddy
- Roll No : A24126552078
- Department : CSM
- CGPA : 8.88

Screenshot 3: CGPA Input Validation Alert

Client-side JavaScript validation triggering an alert popup when an invalid CGPA value is entered.



8. KEY CODE EXPLANATION

- `class Student` — Defines the Student class and its four properties. The constructor assigns values supplied during instantiation.
- `document.getElementById()` — Selects existing HTML input fields, button, and profile container.
- `addEventListener('click', ...)` — Attaches event handler to execute profile generation code on button click.
- `new Student(...)` — Instantiates a Student object with form values.
- `document.createElement()` & `textContent` — Dynamically builds HTML elements and assigns text content safely.
- `appendChild()` — Appends dynamic elements to the DOM.

9. LEARNING OUTCOMES

- Understand JavaScript class structure and object instantiation.
- Master DOM selection, dynamic element creation, and DOM manipulation.
- Implement client-side form validation using regex and numeric bounds.
- Handle browser events and dynamically apply CSS styles.

10. GITHUB SUBMISSION

The final source code is uploaded to GitHub:

https://github.com/venugopalreddychittela/full-stack-web-development/tree/main/01-Observation/obsTask2_student_profile